

Project: 6 On Six

A Reimagined Volleyball Learning Application

Final Report

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By including our names on the cover, we acknowledge that the work is our own and that it was done at McMaster. We recognize that printing our names on the electronic copy is equivalent to signing our names on the document.

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Abstract

This project addresses a significant need in volleyball education, where comprehensive knowledge of basic concepts, such as rotational systems, is missing among players across all levels. Players lack spatial awareness and practical application in volleyball because learning resources are often fragmented, non-interactive, inaccessible, or very expensive. To address this issue, we are creating "6 on Six," a mobile application that brings Duolingo's successful gamified microlearning approach to volleyball instruction. Our solution utilizes Flutter for cross-platform mobile application development, Firebase for backend services, and PostgreSQL for database management to develop an offline-enabled application with interactive lessons, increasing difficulty, and engagement mechanisms. The project has reached the design stage, specifying technology selection, architectural design, and implementation planning. Upon completion, 6 on Six will lead a new generation of sports learning apps that bridge the gap between free yet unstructured alternatives and high-cost professional analysis software by offering low-cost, engaging, and effective volleyball learning that optimizes knowledge retention using proven gamification methods.

Executive Summary

Many volleyball players, from novices to professionals, have small knowledge gaps, including but not limited to terminology, strategies, and, more importantly, rotations. A lack of rotational knowledge may come in the form of only knowing the rotation relevant to you or knowing none at all. This stems from a mix of positional bias, desire to play at higher levels early, and even outright negligence due to complexity.

We plan to create a mobile application called 6 On Six to address these issues. The goal of 6 On Six is to bridge the gap between a working knowledge of volleyball and players who wish to learn more about the sport. Apart from other online resources that exist to satisfy these needs, such as Coach Donny's "Elevate Yourself" YouTube channel or the learnvolley.com website, 6 On Six aims to deliver a medium for learning new concepts that has already seen substantial success, which is to leverage the language learning model of a mobile application called Duolingo.

The application's design will feature a layout that resembles Duolingo's. More specifically, it will be a quiz-based learning structure that deploys modules and sectioned areas of learning, all of which are taught via simple examples such as multiple-choice, fill-in-the-blank, and drag-and-drop questions.

The ideal outcome would be for participants/users to complete various modules in a section and then complete a unit quiz to pass the entire section. This way, users can compartmentalize their learning and retain knowledge through part-by-part recall rather than simultaneously delivering substantial content. More importantly, users could track their progress and be visually informed of how far they've come in learning the game of volleyball and what precisely they do or do not know.

Gamification, ease of accessibility, and effective learning techniques are a large part of what makes Duolingo's learning model a huge success. 6 On Six aims to create an app that follows in these footsteps. The benefit of following this model is already proven, but in scrutiny, applying subtle pressure to encourage users to continue learning is what most other resources are missing out on, and is also what Duolingo has mastered. If someone desires to learn but falls short, it is expected that their learning stops there and will not likely pick up for a while. On the other hand, Duolingo takes advantage of its users' initial drive to start learning, and ensures that they stay committed to their goals.

As straightforward as the plan may sound, the biggest challenge is the absence of preexisting knowledge or experience in mobile app development. What we currently experience from Duolingo is primarily the front-end user interface. We can logically understand why the Duolingo learning model works at a surface level. Still, we don't know much about what happens behind the scenes or how Duolingo knows precisely what the users should be exposed to to secure their commitment to learning. This is likely our most significant hurdle in creating an app that meets the minimum functional requirements of enabling teaching and retaining consistent participation.

But, in a perfect world where all goes well, our immediate next step in future implementation is to include an online component which allows users to create their profiles and accounts. This would be followed by social features such as leaderboards, two-way learning streaks, friendly duels, etc.

Introduction

Motivation

Now more than ever, as the sport grows, there is a systematic gap in volleyball knowledge among players of all levels. With physical skills being a primary emphasis in the sport from a young age, certain fundamental game mechanics (with a focus on rotational systems) tend to be either underdeveloped or missing altogether. This mandates a comprehensive, reliable, and effective educational tool to address the incomplete and/or fragmented understanding of volleyball concepts and skill application.

How the problem manifests can be evident at many stages of a volleyball athlete's career and in many different ways. The most identifiable of these is often present during in-game participation, but is also common in practices. Symptoms usually include confusion with starting locations, difficulty adapting to different positions, collisions/abandoned spots in defensive plays, and an inability to implement more complex strategies due to weak foundational knowledge.

The result of these problems and their visible symptoms also comes in various negative drawbacks. At an individual level, the most obvious negative result is stunted development. If your tactical, in-game understanding hits the ceiling first, you cannot rely on physical skills to complete your development. But, at a team level, things get even worse. Not only is there a blockage of strategy, but practice time (where players SHOULD be focusing on physical skill development) also becomes inefficient. Even in the case of more casual and recreational settings, teams will experience erratic results from a core breakdown of positional knowledge. Ultimately, the potential for both great athletic competence and satisfaction in the game, at all levels, will be short-sighted when there are systematic gaps in volleyball knowledge.

Problem Statement

There is a critical education gap in the volleyball community where players at all levels, from recreational to professional, have incomplete knowledge of fundamental concepts, particularly rotational systems. This is due to positional bias (players only learning their specific position), being rushed through skill levels without adequate base training, and avoiding higher-level tactical elements due to complexity. These gaps create noticeable performance issues during games, cap player development potential, and frustrate players and coaches. The "6 on Six" mobile application addresses this industry-wide problem by providing an interactive, structured learning tool that teaches essential volleyball basics like positioning, rotations, terminology, and strategy to enable more comprehensive player development regardless of experience level or positional specialization.

Project Objective and Requirements

Objective	Requirement
Develop an interactive mobile application, styled after Duolingo, to educate volleyball players on essential skills, emphasizing rotations.	100% of core education content must be accessible offline.
Create a structured interface to guide users through a progressive volleyball-concepts learning model.	At least 80 quiz questions in total with an instant feedback mechanism (i.e. confirm if answer is right or wrong, and if incorrect, show correct answer).
Ensure accessibility to players of all skill levels to promote continuous development and retention of volleyball knowledge.	Minimum of 10 modules covering rotations, terminology, and strategies.

Establish a way to monitor and track user progress.	Cannot consume more than 200 MB of storage space during installation.
The design must allow for a framework that can scale with changing volleyball rules and/or techniques.	Glossary must include at least 50 volleyball terms with search or alphabetically orderable functionality.

Literature Review

Duolingo's Learning Model

Duolingo revolutionized learning languages using its smartphone app, providing a model that can be replicated in other fields of learning, including sports. In their study (Settles and Meeder, 2016), they show how Duolingo's learning model, which uses spaced repetition, maximizes knowledge absorption and retention by predicting when it is likely to be forgotten and reintroducing key concepts as a review section in an active module. By implementing this core mechanic of showing users key content more than once, they maintained a 95% retention rate instead of standard learning, which retains 75-80%.

Fundamentally, Duolingo employs features such as microlearning sessions (5-minute lessons), gamification elements (e.g., streaks, leaderboards, achievements, etc.), progressive difficulties to omit redundancy in areas of high competency, and, most importantly, immediate feedback that provides users with an assessment of their responses as soon as they finish a section. (Hamari et al., 2014)

While Duolingo has seen evident success in providing an effective medium for learning languages to over 500 million global users (Duolingo, 2023), its model has not been thoroughly applied in sports education. However, this is not to say that Duolingo hasn't branched into other areas of education. They've also explored units in math and music, up to around an intermediate level, so it could also be a matter of business planning and supplying the demand of what's most desired in their current user base. The company's incremental and methodical approach could translate well for teaching volleyball knowledge, but would likely require adaptation to accommodate spatial-physical concepts instead of linguistic ones.

Sports Education Apps Today

Unlike modern language learning apps, which carry a wide range of comprehensive digital learning materials, sports training apps are relatively underdeveloped. According to a systematic review by (Rossi et al. 2020), most sports training apps focus on physical fitness rather than conceptual or tactical awareness.

HomeCourt, a basketball training app that analyzes shots and is driven by AI, is one of the more advanced sport education technologies (Kamenetzky, 2019). Its emphasis, though, remains on the performance of technical skills rather than on game mechanics conceptual knowledge.

The absence of current models of sports education apps is both a challenge and an opportunity for 6 on Six. While we cannot draw upon pre-existing, in-place models, we also get the chance to innovate by integrating Duolingo's proven engagement drivers and volleyball-focused content.

Volleyball-Specific Resources

Elevate Yourself (YouTube)

Elevate Yourself, a YouTube channel hosted by Coach Donny, is one of the most recognized educational resources on volleyball today. It is subscribed to by 600,000 people and has 78 million views (as of 2025). His content consists of extensive tutorials on rotations, tactics, and movement in volleyball. The platform's benefits include visual demonstrations of movement and correct technique, comprehensive explanations of tactical concepts by a tenured coach, free availability to any viewer, and detailed discussion of principles of volleyball. However, some weaknesses of the platform include non-interactive presentation of linear material that cannot cater to the viewer's needs, occasionally poorly organized presentation of concepts in sequence, and lack of a knowledge retention test system. Generally, it lacks proper engagement tools to ensure regular learning and offline access without a premium YouTube subscription. Coach Donny's program bears testament to the broad interest in learning volleyball, but exposes the need for interactive and systematic learning modes that cater to specific learning needs.

Learnvolley.com (Website)

Learnvolley.com provides interactive court position diagrams in various scenarios (i.e. serve and receive or defence), role-specific coaching per role, and simple player movement pattern animations. While these directly address the visualization problem of rotations, the web-based system does have significant shortcomings. With limited elements of gamification and engagement features, it's easy to get lost in the sea of content and can feel overwhelming. The website also lacks tracking of the user’s learning, so even with their visualized and animated delivery of knowledge, much of what's presented to the user cannot guarantee their retention. There is also an absence of built-in examination or testing functionality and little offline capability, making it incompatible with certain volleyball practice environments where internet access cannot be ensured. Although learnvolley.com addresses the specific problem of rotation visualization, it does not supply the educational structure and engagement mechanism needed for lasting learning and retention.

Gaps in Existing Solutions

Reviewing existing volleyball educational resources unmasks specific gaps that the 6 on Six app is built to address. Current volleyball resources used to learn terminology, strategies, and rotations are either in the form of non-interactive video tutorials (YouTube) or web interfaces incompatible for application in the practice environment. This results in a void of potential interactive learning in the mobile environment. Like Duolingo, no existing volleyball resources implement game-like engagement features critical to support consistent, long-term engagement in learning materials, resulting in decreased retention and inconsistent usage patterns. Advanced analysis tools are too expensive for amateur players, and free materials never have structured paths of instruction and full contents, presenting access barriers that confine volleyball instruction to professionals or well-motivated self-learners. Most web-based resources in volleyball also demand consistent internet access, limiting them in the gym environment where there is sometimes spotty or unreliable internet access.

Positioning of 6 On Six

Our 6 on Six app does not attempt to compete head-to-head with professional-grade volleyball analysis tools that cost thousands of dollars and offer professional-grade performance analysis. Instead, we've designed our solution to bridge the learning gap for individual athletes who need to gain the fundamentals of volleyball, particularly terminology, strategies, and rotations, all in an affordable mobile environment. Drawing inspiration from Duolingo's successful learning-engagement model and adapting it to the volleyball learning environment, we believe 6 on Six can surpass other volleyball learning tools in learning value without a significant cost or unaffordability. The application will be the first to merge incremental learning paths and game-like engagement features to drive it, a mobile-first design to learn anywhere, and offline capability for in-gym use. We plan to design the core product to deliver interactive tests and quizzes to facilitate retention of learning, with easy module completion tracking.

While we recognize that our teaching style can't aspire to the analytic depth of enterprise-class volleyball software, I'd like 6 on Six to design a new sports learning app category that prioritizes conceptual comprehension and retention through proven engagement mechanisms. Future app versions may support increasingly complex personalization functions and adaptive learning rules. Still, in this initial version, we'd like to give a firm grounding of volleyball knowledge through a compelling, easy-to-use interface.

Design

Alternative Solutions

While designing the 6 on Six volleyball learning app, we evaluated numerous platforms and learning model opportunities to most effectively meet volleyball knowledge needs.

We balanced web vs. mobile app options on the platform. Web apps can be accessed on any device, are easier to build, and are updated in real time without going through the approval of app stores. However, they would bring limited offline capability as well as limited mobile device performance. On the other hand, a mobile app delivers better performance, full offline capability for use in gyms, push functionality for driving higher retention, and access to native device capabilities.

After carefully evaluating the options, it became clear that the mobile solution using Flutter would be more appropriate for our learning objectives. Offline support is particularly suitable for users in sports centers with poor connectivity, whereas the ability to push notifications is a significant benefit for learning continuity and retainability of users.

For the learning platform, we considered video learning, quizzing in the traditional form, interactive tutorials, and gamified microlearning in Duolingo format. Video learning is excellent for visual descriptions but is passive in nature. Traditional quizzing directly assesses knowledge but without any interactivity. Interactive tutorials provide learning and practice in combination but can be overwhelming for beginners.

The gamified microlearning model stood out due to its brief, focused learning sessions, spaced repetition for better retention, built-in motivation systems, and variable difficulty incrementation. Research predicted this model would increase retention up to about 95 percent compared to 75-80 percent in conventional means. The success of such a model in language learning sites like Duolingo had significant potential for its applicability in teaching volleyball.

For designing the interface, we considered court-based designs (visualization of the court), position-based structure (content according to player position), and progressive learning paths (linear progression in terms of concepts). We used the progressive learning path as our overall structure in filling knowledge gaps in sequential order and, ideally, we would incorporate court visualizations as components in specific modules instead of the overall design emphasis.

Proposed Solution

Our product is a smartphone learning software that converts the Duolingo learning model into principles of volleyball, with rotation comprehension as the focus. It is a three-tier architecture split into three related but distinct functional components:

- **Frontend (User Interface):** We've opted for Flutter, an open-source User Interface (UI) software development kit built by Google that allows us to build natively compiled applications for mobile, web, and desktop from just one shared codebase. Flutter is written in the Dart programming language and features rich, customizable widgets that assist in responsive, high-quality interfaces. For non-developers, we're doing the iOS and Android versions simultaneously, cutting down development times significantly and ensuring a uniform experience across devices.

- **Backend (Server Logic):** For backend systems, we are leveraging Firebase, Google's full development platform, with pre-existing solutions for authentication, cloud storage, and real-time database support. Firebase manages the "behind-the-scenes" aspects such as user management, data storage, and synchronization without us having to implement complicated server infrastructure.

- **Database (Data Storage):** We've selected PostgreSQL (often called "Postgres"), a powerful open-source relational database system with over 30 years of active development. PostgreSQL excels at storing structured data and performing advanced queries, so it's ideal for holding our educational content and user progress information.

The interface has a contemporary look with volleyball-themed graphics. The home page has a learning pathway presented as a progression tree, with lessons divided by level of difficulty as well as type of topic. Each learning module contains 5-8 small lessons that can be completed in 3-5 minutes, consisting primarily of multiple choice questions, drag-and-drop position elements, and terminology (fill in the blank). Visualization of progress through indicators of completeness, as well as achievement in terms of streaks and overall completion, assists in keeping users motivated for ongoing participation.

Material is presented in sequential order, beginning with the fundamentals (foundational court positions, terms, roles for the players, and rules), rotational fundamentals (the six basic rotational positions and serving order rules), intermediate modules on rotational movement for setter-anchored movement between offence and defence. More advanced modules cover 5-1 and 6-2 rotational systems, scenario-based position switches, and final chapters on rotational game situations with solutions for particular circumstances and tactical adjustments.

The app uses a system of testing that is difficult but within reach. Users are granted up to 10 errors in 24 hours before they need to wait or use premium options, giving them a sense of responsibility without losing enthusiasm. Users must complete each learning unit by completing all previous units in the sequence and scoring 100% on the review quiz, with two daily attempts (guided by the error limit). Suppose users are unsuccessful in completing a unit within their allocation of attempts. In that case, they can reattempt it after waiting for 1 hour or use premium options to attempt it again immediately.

After initiating a question module, a user cannot cancel without losing progress on the module, motivating finished work and concentration. Multiple choice with simple graphical elements, drag-and-drop placement questions, recognition of correct rotational motions, terminology pairs, and scenario problems are among the question formats used.

To support interaction and retention, the app features some gamification elements such as a learning day streak system, achieving badges for the accomplishment of certain tasks, visual representation of progress, level-ups for mastery of volleyball knowledge, and reminders for practice.

The glossary is an educational guide and resource comprising in-depth volleyball terminology containing complete definitions of terms, player positions, tactics, and rules. Where appropriate, terms are included with diagrams or sketches and instant access to specific terms through a search functionality. Examples include:

- **5-1 Rotation:** In this system, one player takes on only one position on the court, having one setter and five different attackers (middle blockers, outside hitters, opposite).

- **Setter (S):** This is the position in which the player is in charge of the second ball contact and "sets up" their attackers for the third contact (attack).

- **Float Serve:** A serve where the ball is hit with open-hand contact so that it clears the net without any spin and "floats" as it flies in on the passers.

An app that is enabled for offline use as soon as it is downloaded, with learning content cached on the device locally and progress stored locally with automatic syncing on connection ensures that it is accessible in regions of poor net connectivity in the gym, for example.

Tech Stack Justification

Our technology stack is well suited to this learning app for the following reasons:

Cross-Platform Performance: Using Flutter, we can build for both Android and iOS from the same codebase, which is essential to targeting the broadest range of volleyball players independent of device preference. Unlike building two native applications, Flutter reduces development time by about 40% without sacrificing near-native performance.

Offline Capability: The combination of Flutter's offline data storage feature and Firebase's offline data syncing feature enables users to use it smoothly even in areas with poor connectivity. Firebase automatically syncs the progress data when the users are connected again without requiring manual intervention.

Scalability: PostgreSQL can efficiently manage huge amounts of educational material and user data as our user base increases. Its JSONB (JavaScript Object Notation Binary) support is especially useful in storing quiz structures and learning modules that could change over time.

Development Accessibility: For a small development team, this stack offers the appropriate balance between power and ease of use. Firebase eliminates backend complications that would otherwise require committed server infrastructure and database administration.

Cost Effectiveness: The stack utilizes several free open-source components, maintaining low initial development costs while offering enterprise-class functionality.

The primary drawback of our chosen stack is the Dart learning curve (Flutter's programming language) for programmers familiar with other languages. However, this drawback is balanced by Flutter's well-documented nature and increasing community support for Flutter development.

Although the first release will concentrate on core functionality, the architecture will be built to support future additions such as online capability (user accounts, leaderboards, community features), advanced visualization technology, and coach integration tools. Our technology decisions' modularity will enable these features to be added in phases without necessitating an entire application redesign.

Bill of Materials

Category	Essential Items	Cost Estimate (CAD)	Notes	Links
Development Tools	Visual Studio Code, Flutter SDK, Firebase (Spark Plan), PostgreSQL	$0.00	Already using VS code, other tools are free	Visual Studio Code Install | Flutter Firebase Supabase (Free Tier for PostgreSQL)
Cloud Services	Google Cloud (Firebase hosting)	$0.00	Using existing free tier account	Free Trial and Free Tier Services and Products | Google Cloud
Hardware	Development machines, iPad Pro for design, mobile test devices	$0.00	Already owned equipment	N/A
Design Resources	Procreate (for iPad)	$17.99 (one-time cost)	Using existing iPad Pro for graphics work	Procreate
Deployment	Apple Developer Account, Google Play Developer Account	$140.00	Required for app publication	Apple Developer Program Google Play for business | Launch & monetize your apps | Google Play Console
Total Initial Investment		$157.99		

Completed Work

The work we completed mainly consists of this capstone project's blueprint. We finalized the technology stack that we would be working with, the integration of the technology, the cost of usage and the details of each part of our project plan for capstone II.

Database selection:

PostgreSQL was selected for this project because it is highly scalable, highly flexible, and has robust data-handling capabilities, making it more suitable than the older-style databases like MySQL. The application requires monitoring progress of users, managing dynamic lesson data, and supporting multiple user roles—requirements that are serviced well by PostgreSQL through its efficient query processing and performance optimization of large sets of read-heavy data. Its JSONB support offers adaptive, structured storage for items like quizzes, drills, and lesson modules independent of the confines of a hard schema, which the app requires with its constantly updating and customizable nature. Additionally, the ability of PostgreSQL to be entirely integrated with newer frameworks and cloud services enables flawless deployment and upkeep.

While performance is crucial for PostgreSQL, the database also has features that enhance security and administration for the app. Its strong permission and role system facilitates easy enforcement of user privilege levels, solidifying secure boundaries between administrators, moderators, and normal users. Full ACID compliance offers dependability for activities such as user authentication, progress tracking, and lesson updates. Postgres stands to benefit from MySQL in that it provides better concurrency control, more advanced query support, and extensibility—which allows custom functions, triggers, and future data models as the application expands. Through the use of PostgreSQL, we are implementing a future-proof, secure, and efficient backend solution that is consistent with the needs of a modern, interactive volleyball training site.

Factor	***PostgreSQL***	***MySQL***
JSON Handling	Advanced (JSONB support)	Basic (limited JSON functions)
Query Complexity	Excellent for complex queries	Moderate, best for simple queries
Role Management	Fine-grained and flexible	Basic and limited
Performance (Read-heavy)	Optimized for large, read-heavy workloads	Fast for lightweight operations
Schema Flexibility	High – supports dynamic, semi-structured data	Moderate – rigid schema structure
Ease of Setup	Moderate – requires configuration	Easy – fast and simple to deploy

Front-End selection:

For our front-end implementation of the volleyball training application, we considered having two out-shining plans in place: one web-focused approach based on HTML, CSS, and JavaScript and another mobile-centric approach using Flutter. Each had specific positives and negatives pertaining to accessibility, performance, usability, development and long-term maintainability. The app being centered on interactive lessons, quizzes, tracking progress, and cross-device responsive interaction made it essential to vet each of the technologies in regard to being able to sustain these fundamental capabilities.

Web-based is a highly robust accessibility model in which users can execute the app inside any modern web browser without installing it. It's a simple method to maintain things current since updates can be performed server-side without involving the users. Additionally, React.js and Vue.js libraries support rapid development cycles and dynamic UIs. However, web apps are typically internet dependent, offer limited offline capabilities, and are unable to take advantage of native device features like motion sensors, cameras, and GPS. These can be a hindrance to the user experience, especially for a performance-based training app that requires responsiveness and interactivity.

After weighing both frontend possibilities, we decided to develop our app using Flutter. While HTML is easier to apply and more available, Flutter offers a much more immersive and high-quality feature set. Its ability to natively compile offers smooth animation and responsive interaction, as well as complete support for device hardware. Most significantly, Flutter supports offline functionality, and users are able to see lessons, drills, and progress tracking even when they are offline. Adding push notifications and higher engagement features aids in user retention. Although Flutter development is a bit more effort initially and to sustain, its performance, feature list, and scalability in the future make it the ideal solution for achieving our volleyball training app goals.

Factor	***HTML (Web-Based)***	***Flutter (Mobile App)***
Cross-Platform Support	Works on all devices with a browser	Works on both iOS and Android
Performance	Lower on mobile, browser-dependent	High-performance, native-like
Offline Access	Limited, unless using PWAs	Full offline functionality
Ease of Updates	Instant updates via server	Requires app store approvals
User Engagement	Lower retention, no push notifications	Higher retention with push notifications
Development Complexity	Easier, wide developer availability	More complex, requires Flutter/Dart expertise
Device Feature Integration	Limited, browser-based APIs	Full access to camera, sensors, GPS, etc.

Back-End Selection:

We used Firebase to create the backend for our volley practice application because it plays well with Flutter, has real-time data, and scales well. Firebase Cloud Firestore and Realtime Database give us live updates for monitoring progress, leaderboards, and interactive practice, which are the dynamic features of the application. Real-time feature enhances user experience as it updates in real time whenever data changes without requiring users to refresh the app. With the interactivity of our app, a primary concern was having a responsive and smooth user experience, which Firebase allows.

The second major advantage of Firebase is that it supports Flutter natively, which makes it simple to integrate major backend features such as user authentication, data storage, and push notifications. Both being Google technologies, they are highly optimized to play nicely together, with less compatibility headache and quicker development. Firebase also offers out-of-the-box scalability with Google Cloud infrastructure, allowing our app to scale seamlessly. No matter whether it's serving hundreds or hundreds of thousands of users, Firebase will scale seamlessly without loss of performance and is a rock-solid foundation on which the app can grow.

Additionally, Firebase offers offline access, by which users are able to view training plans and lessons even in areas with weak internet connectivity or no connectivity. This is extremely convenient for athletes who train in variable or remote conditions. Firebase Authentication offers various secure login methods, including email and Google Sign-In, that ensure data security and confidentiality of the users. Backend activities such as handling analytics processing or notification sending are done through Firebase Cloud Functions without the need for an external server infrastructure. We can send targeted push messages using Firebase Cloud Messaging to optimize user engagement and retention. Firebase Hosting and native analytics features also assist in straightforward deployment and monitoring of user activity, enabling us to build and optimize the app incrementally. Overall, Firebase provides a strong, yet malleable backend option that enables us to concentrate on providing a good user experience without having to give attention to intricate server infrastructure.

Discussion

Viability

Both of us believe that the 6 On Six initiative provides a very practical answer to a prevailing problem in the world of volleyball; a lack of elementary understanding in subjects ranging from rotations, vocabulary, to positions. Our research led us to notice that current materials available, such as video lessons and web-based static websites, are not providing the organized, interactive, and inspirational materials that assure long-term learning retention. With Duolingo's successful gamified microlearning platform model, we believe 6 On Six is well-positioned to bridge the learning gap among volleyball players of all levels.

Technologically, our selection of technology stack positions the application for success. By leveraging Flutter, we will have the capability to develop for the Android and iOS platforms from a single codebase, saving time without sacrificing high-performance apps that are native-like. Flutter support for offline modes is especially important in sport environments where network availability is never assured. Our Firebase-backend backend simplifies the most significant processes such as user authentication, real-time data synchronization, and cloud storage. PostgreSQL as a highly scalable and flexible platform for dynamic learning content management and monitoring user progress. Collectively, these technologies give us a strong foundation to support both existing and future versions of the app.

Cost-wise, we have kept the project highly cost-effective. With an initial investment of mere CAD 157.99, we've utilized free development tools, existing hardware, and free plans of cloud services. This allows us to focus our energies on functionality and user experience without the cost. More importantly, the technologies that we've utilized are modular and scalable, i.e., we can continue to develop the app in subsequent phases without having to redesign everything.

One of the most exciting aspects of 6 On Six is the opportunity to engage users in a way that traditional volleyball learning tools do not. By incorporating features such as streaks, badges, daily goals, and progress tracking, we're creating a system that not only educates but motivates. The microlearning structure encourages consistency and retention while making the experience enjoyable and accessible. This approach has proven effective in language learning and, with suitable modification, holds much promise in sports learning.

Of course, we are aware that there are obstacles to be overcome. As a team, our experience in mobile app development is minimal, and we understand that to take a model such as Duolingo's and apply it to a spatial-physical setting such as volleyball will require careful design and user testing. However, we have accounted for this by having a phased rollout and placing core functionality first on the agenda for the launch. With a solid master plan, a technology stack that scales, and a clear plan for future upgrades, we are certain that we can overcome these issues.

In summary, we believe that 6 On Six is a viable and revolutionary product that can make a tangible impact with the volleyball community. By combining interactive, immersive learning pedagogies with mobile accessibility and offline functionality, our app stands out from existing solutions. With more development, consumer feedback, and strategic iteration, we're truly excited to see what this product will become.

Future Challenges and Contingency Planning

As we embark on the development stage of 6 On Six, our team is eager to encounter a few challenges that can impact both the efficiency and speed of our work. One of the most pressing logistical issues is our off-sequence academic calendars. While teams working in the normal stream of academics share similar loads of courses and co-op projects across terms, our group members are working with varying loads of courses and co-op projects in varying terms. This makes windows of overlapping availability and compounds synchronous collaboration more difficult, especially when the work involves tasks such as feature integration, interface testing, and code review. Moreover, with some of us starting co-op terms mid-Capstone II, there is the added complexity of balancing full-time work commitments with ongoing project commitments.

Aside from scheduling, a potential issue is the learning curve on our tech stack of choice. Although Flutter and Firebase are ideal companions in terms of scalability and offline features we require, there are individuals among us who are also learning Dart and getting to grips with backend design in Firebase. Because of this, there can be moments where development falls behind as we learn, integrate, and test major features extensively.

To prevent such problems in the first place, we've set up several contingency plans. For one, we're developing the app modually, which means that we're both individually developing isolated segments of the app that eventually come together. That reduces dependencies on simultaneous real-time collaboration and offers us greater latitude to schedule independent pieces of work. We will also employ tools like GitHub Projects and team Kanban boards to monitor progress asynchronously, while leaving ourselves open to one another's schedules.

Secondly, as preliminary to the technical challenges, we've already begun developing learning materials and onboarding documents within the group. These will have the effect of ensuring that any specific member can fill in for another when necessary so workload is distributed evenly even during high-demand periods such as midterms or co-op rotations.

Lastly, we've also committed to keeping in contact with one another constantly with weekly asynchronous progress updates and monthly scheduled check-ins where everyone has the opportunity to synchronize on progress, blockers, and upcoming milestones. Where some deliverables cannot be done within the initial timeline due to external factors, we've also scheduled buffer weeks towards the latter part of Capstone II so that there is catch-up work, bug fixing, and testing.

In short, while we do recognize that our staggered school schedules and limited earlier experience with some of the tools present humongous challenges, we've made our development cycle robust and resilient. With good communication strategies, modular design, and flexibility time considered, we feel confident that our team can deliver a working and useful version of 6 On Six by the last capstone cycle.

Project Plan (Capstone II)

Duration: 12 Weeks

Week	Milestone	Tasks & Deliverables	Lead(s)	Notes
Week 1	**Project Kickoff & Task Assignment**	- Re-align team goals - Assign modules based on expertise/availability - Finalize development timeline - Set up GitHub repo, Kanban board	All	Sync up after co-op breaks
Week 2	**Backend Setup**	- Firebase project initialization - PostgreSQL schema design finalized - Implement user auth (email sign-in)	Backend Lead	Minimal UI needed this week
Week 3	**Frontend Base & Navigation**	- Build core Flutter UI layout - Implement main navigation (Home, Modules, Glossary, Profile)	Frontend Lead	Modular work begins

Week 4	**Lesson Module Framework**	- Create module structure (data schema) - Build placeholder for quiz templates - Display mock content	UI/UX + Backend	Start with 1 sample module
Week 5	**Quiz Engine**	- Build logic for MCQ, drag-and-drop, and fill-in-the-blank - Add answer feedback flow	Backend + Frontend	Reuse template for later modules
Week 6	**Progress Tracking + Gamification v1**	- Implement basic progress tracking - Add streak counter, badges (static logic)	Frontend	Focus on logic, visuals later
Week 7	**Glossary Module**	- Develop searchable glossary - Include volleyball terms + visuals	Backend + Frontend	Should work offline
Week 8	**Offline Mode Implementation**	- Cache data locally - Ensure lessons accessible without internet	Backend Lead	Test in gym-like environments
Week 9	**Unit Completion & Review Quiz Flow**	- Implement locking/unlocking mechanism - Add review quizzes & retry limit logic	Full Stack	Finalize module experience
Week 10	**Polish & Visual Enhancements**	- UI styling + visual feedback - Animated progress tracker - Add volleyball-themed assets	UI/UX	Prioritize usability
Week 11	**Internal Testing & Bug Fixes**	- Team-wide app testing - Log bugs & patch issues - Optimize performance	All	Allocate time for major fixes
Week 12	**Final Report & Demo Prep**	- Record screen demo - Finalize project presentation - Submit updated documentation	All	Dry-run of presentation included
Buffer Week(s)	**Catch-up & Final Touches**	- Finish pending features - Fix unresolved issues - Final polish and deployment	All	Use only if needed

Conclusion

The development of 6 On Six has been driven by a straightforward yet meaningful mission: to bridge the learning gap in volleyball by offering a mobile-first, interactive learning experience that is accessible, engaging, and effective. Through research, design planning, and technical analysis, our team has laid the foundation for a revolutionary app that leverages successful learning models like Duolingo's gamified microlearning system to the particular requirements of sports education.

Our decisions around technology, design structure, and user engagement have been made carefully, considering both feasibility and long-term value. By leveraging Flutter, Firebase, and PostgreSQL tools, we've created a scalable and cost-effective architecture that supports offline functionality, real-time interaction, and an intuitive user experience. These choices allow us to tackle the short-term learning needs of volleyball players with the flexibility to incorporate possible future features such as coach integration, leaderboards, and advanced learning modules.

As we proceed to Capstone II, we know that challenges lie ahead, including asynchronous team schedules, co-op activities, and technical complexity to develop a trustworthy, high-performance mobile app. Yet, with prior planning, the explicit allocation of tasks and requirements, and the inclusion of collaborative software tools, we have no questions in our production of a working minimum viable product (MVP) that realizes the value and functionalities central to 6 On Six.

Ultimately, we believe that 6 On Six can leave a mark on learning volleyball. With its structured, easy-to-access content coupled with the mechanisms of encouragement that propel users forward, our app is not only destined to teach but also to incentivize constant learning and development in players across the board. We are eagerly awaiting the opportunity to bring this important project forward and cultivate it throughout the upcoming term.

References

Realtime Database Documentation." *Firebase*, Firebase, https://firebase.google.com/docs/database.
"Firebase Hosting Overview." *Firebase*, Firebase, https://firebase.google.com/docs/hosting.
"Flutter Documentation." *Flutter*, Google, https://flutter.dev/docs.
"Flutter Performance Overview." *Flutter*, Google, https://flutter.dev/docs/perf.
"Using Firebase with Flutter." *FlutterFire*, https://firebase.flutter.dev/docs/overview.
"Flutter Push Notifications Guide." *FlutterFire*, https://firebase.flutter.dev/docs/messaging/overview.
"PostgreSQL Documentation." *PostgreSQL*, The PostgreSQL Global Development Group, https://www.postgresql.org/docs/.
"Advantages of PostgreSQL." *PostgreSQL*, The PostgreSQL Global Development Group, https://www.postgresql.org/about/.
"HTML Introduction." *MDN Web Docs*, Mozilla, https://developer.mozilla.org/en-US/docs/Web/HTML.
"JavaScript Documentation." *MDN Web Docs*, Mozilla, https://developer.mozilla.org/en-US/docs/Web/JavaScript.
"CSS Documentation." *MDN Web Docs*, Mozilla, https://developer.mozilla.org/en-US/docs/Web/CSS.
"Progressive Web Apps Overview." *MDN Web Docs*, Mozilla, https://developer.mozilla.org/en-US/docs/Web/Progressive_web_apps.
"Firebase vs SQL Databases." *MongoDB*, MongoDB, https://www.mongodb.com/compare/mongodb-vs-firebase.
"Why You Should Choose Firebase as Your Backend for Your Mobile App." *Medium*, Firebase Developers, https://medium.com/firebase-developers/why-you-should-choose-firebase-as-your-backend-for-your-mobile-app-9d6d4296eae3.
"Flutter for Web." *Flutter*, Google, https://flutter.dev/web.
"Web vs Mobile: Which Is Best for Your Project?" *CreativeBloq*, https://www.creativebloq.com/inspiration/web-vs-mobile-which-is-best-for-your-project.
Duolingo. (2023). About Us - Duolingo. https://www.duolingo.com/info
Gold Volleyball. (2022). Data Volley 4 Professional. https://goldvolleyball.com/product/data-volley-4-professional/
Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does gamification work?—A literature review of empirical studies on gamification. 47th Hawaii International Conference on System Sciences, 3025-3034.
Kamenetzky, A. (2019). How HomeCourt, an AI-powered app, is changing basketball. The Athletic. https://theathletic.com/989033/2019/05/28/how-homecourt-an-ai-powered-app-is-changing-basketball/
Rossi, A., Pappalardo, L., Cintia, P., Iaia, F. M., Fernàndez, J., & Medina, D. (2020). Effective injury prediction in professional soccer with GPS data and machine learning. PLOS ONE, 15(12), e0243843.
Settles, B., & Meeder, B. (2016). A trainable spaced repetition model for language learning. Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics, 1848-1858.